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Base station antenna

Abstract

The present disclosure relates to a base station antenna; the base station antenna comprises: a radio; a feed board and a radiating element array mounted on the feed board, where the radiating element array comprises a plurality of radiating elements, and each radiating element extends forward from the feed board; a radome arranged in front of the radiating element array; and a plurality of radome supporting members mounted on the radio and extending forward from the radio to the radome for supporting the radome.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6069590	12/1999	Thompson, Jr.	343/797	H01Q 21/26
2018/0040948	12/2017	Göttl	N/A	H01Q 19/108
2018/0337443	12/2017	Bisiules	N/A	H01Q 1/246
2021/0075092	12/2020	Su et al.	N/A	N/A
2023/0104131	12/2022	Varnoosfaderani	343/702	H01Q 21/26
2023/0208014	12/2022	Zhang	343/745	H01Q 19/06

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
205992589	12/2016	CN	N/A
206546874	12/2016	CN	N/A

OTHER PUBLICATIONS

“Extended European Search Report corresponding to European Application No. 23160023.0 dated Jul. 20, 2023”. cited by applicant

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Background/Summary

RELATED APPLICATION

(1) The present application claims priority from and the benefit of Chinese Patent Application No. 202210202034.3, filed Mar. 3, 2022, the disclosure of which is hereby incorporated herein by reference in full.

FIELD OF THE INVENTION

(2) The present disclosure generally relates to the field of radio antenna, and more specifically, the present disclosure relates to a base station antenna.

BACKGROUND OF THE INVENTION

(3) Base station antennas usually operate in open areas, and are directly subject to the natural forces of rainstorm, snow, dust, and solar radiation, etc., which reduces antenna accuracy of base station antenna systems, shortens their lifespans and causes their work reliability to be poor. Therefore, there is a need to mount a radome in the base station antenna to protect the base station antenna system from external environmental impact. Therefore, how to ensure the stability of the radome in

a reliable manner and to prevent the antenna system from being damaged by the toppling of the radome is a technical problem that urgently needs to be solved by those of ordinary skill in the art.

SUMMARY OF THE INVENTION

(4) Therefore, the objective of the present disclosure is to provide a base station antenna capable of overcoming at least one drawback in the prior art.

(5) According to a first aspect of the present disclosure, a base station antenna is provided; the base station antenna comprises: a radio; a feed board and a radiating element array mounted on the feed board, where the radiating element array comprises a plurality of radiating elements, and each radiating element extends forward from the feed board; a radome arranged in front of the radiating element array; and a plurality of radome supporting members mounted on the radio and extending forward from the radio to the radome for supporting the radome.

(6) According to a second aspect of the present disclosure, a base station antenna is provided; the base station antenna comprises: a feed board; a radiating element array mounted on the feed board, where the radiating element array comprises a plurality of radiating elements and each radiating element extends forward from the feed board; a radome arranged in front of the radiating element array; radome supporting members for supporting the radome mounted on the radome and extending rearward from the radome.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will be explained in more detail by means of embodiments with reference to the accompanying drawings. The schematic drawings are briefly described as follows:

(2) FIG. 1 shows a schematic perspective view of a base station antenna according to a first embodiment of the present disclosure;

(3) FIG. 2 shows a schematic diagram of the base station antenna in FIG. 1 without a radome;

(4) FIG. 3 shows a partial exemplary diagram of the base station antenna in FIG. 1, and also shows the mounting scheme of radome supporting members in the base station antenna;

(5) FIG. 4 shows a modified design scheme of the radome supporting member in FIG. 3;

(6) FIG. 5 is a schematic perspective view of a base station antenna according to a second embodiment of the present disclosure;

(7) FIG. 6 shows a schematic perspective view of a radome of the base station antenna in FIG. 5.

(8) FIG. 7 shows a partial exemplary diagram of the base station antenna in FIG. 5, and shows the mounting scheme of radome supporting members in the base station antenna;

(9) FIG. 8 shows a modified design scheme of the radome supporting member in FIG. 7.

DETAILED DESCRIPTION OF THE INVENTION

(10) The present disclosure will be described below with reference to the attached drawings, which show several examples of the present disclosure. However, it should be understood that the present disclosure can be presented in many different ways and is not limited to the examples described below. In fact, the examples described below are intended to make the present disclosure more complete and to fully explain the protection scope of the present disclosure to those skilled in the art. It should also be understood that the examples disclosed in the present disclosure may be combined in various ways so as to provide more additional examples.

(11) It should be understood that the terms used herein are only used to describe specific examples, and are not intended to limit the scope of the present disclosure. All terms used herein (including technical terms and scientific terms) have meanings normally understood by those skilled in the art unless otherwise defined. For brevity and/or clarity, well-known functions or structures may not be further described in detail.

(12) As used herein, when an element is said to be “on” another element, “attached” to another

element, “connected” to another element, “coupled” to another element, or “in contact with” another element, etc., the element may be directly on another element, attached to another element, connected to another element, coupled to another element, or in contact with another element, or an intermediate element may be present. In contrast, if an element is described as “directly” “on” another element, “directly attached” to another element, “directly connected” to another element, “directly coupled” to another element or “directly in contact with” another element, there will be no intermediate elements. As used herein, when one feature is arranged “adjacent” to another feature, it may mean that one feature has a part overlapping with the adjacent feature or a part located above or below the adjacent feature.

(13) As used herein, spatial relationship terms such as “upper”, “lower”, “left”, “right”, “front”, “back”, “high” and “low” can explain the relationship between one feature and another in the drawings. It should be understood that, in addition to the orientations shown in the attached drawings, the terms expressing spatial relations also comprise different orientations of a device in use or operation. For example, when a device in the attached drawings rotates reversely, the features originally described as being “below” other features now can be described as being “above” the other features”. The device may also be oriented by other means (rotated by 90 degrees or at other locations), and at this time, a relative spatial relation will be explained accordingly.

(14) As used herein, the term “A or B” comprises “A and B” and “A or B”, not exclusively “A” or “B”, unless otherwise specified.

(15) As used herein, the term “schematic” or “exemplary” means “serving as an example, instance or explanation”, not as a “model” to be accurately copied”. Any realization method described exemplarily herein may not be necessarily interpreted as being preferable or advantageous over other realization methods. Furthermore, the present disclosure is not limited by any expressed or implied theory given in the above technical field, background art, summary of the invention or specific embodiments.

(16) As used herein, the word “basically” means including any minor changes caused by design or manufacturing defects, device or component tolerances, environmental influences, and/or other factors.

(17) In addition, for reference purposes only, “first”, “second” and similar terms may also be used herein, and thus are not intended to be limitative. For example, unless the context clearly indicates, the words “first”, “second” and other such numerical words involving structures or elements do not imply a sequence or order.

(18) It should also be understood that when the term “comprise/include” is used herein, it indicates the presence of the specified feature, entirety, step, operation, unit and/or component, but does not exclude the presence or addition of one or a plurality of other features, steps, operations, units and/or components and/or combinations thereof.

(19) The base station antenna may be mounted on an elevated structure, for example, a base station antenna tower, a telegraph pole, a building, or a water tower, such that the longitudinal axis thereof extends substantially perpendicular to the ground. The base station antenna is usually mounted in a radome that provides environmental protection. In the base station antenna, the radome is a structure that protects the base station antenna system from external environmental impact. In terms of electrical performance, the radome has good electromagnetic wave penetration characteristics, and in terms of mechanical performance, the radome is capable of withstanding the effects of harsh external environments (such as rainstorm, snow, sand and solar radiation). Generally, radome supporting members are additionally mounted in the base station antenna for supporting the radome, thereby further stabilizing the radome and preventing the base station antenna system from being damaged by the toppling of the radome. These radome supporting members are generally mounted on for example, an aluminum reflector of the base station antenna. However, the reliable mounting of each radome supporting member on the reflector is rather labor-intensive.

(20) However, the inventor(s) also found that: in some application scenarios, for example, some

low cost and/or low-weight base station antennas are not mounted with a special reflector to reduce the cost and/or weight of the base station antenna. To this end, how to reliably mount radome supporting members for such base station antennas is also a technical problem that urgently needs to be solved.

(21) The base station antenna according to some embodiments of the present disclosure may relate to such reflector-free base station antennas.

(22) The base station antenna according to some embodiments of the present disclosure may be constructed as an active antenna device. The active antenna device may include one or more arrays of radiating elements that operate under fifth generation (5G or higher version) cellular network standards. In 5G mobile communication, the frequency range of communication includes a main frequency band (specific portion of the range 450 MHz-6 GHz) and an extended frequency band (24 GHz-73 GHz, i.e., millimeter wave frequency band, mainly 28 GHz, 39 GHz, 60 GHz and 73 GHz). The frequency range used in 5G mobile communication includes frequency bands that use higher frequencies in the previous generations of mobile communication. These arrays typically have individual amplitude and phase control over subsets of the radiating elements therein and perform active beamforming.

(23) Next, refer to FIGS. 1 to 3, which describe in detail a base station antenna **100** according to a first embodiment of the present disclosure. FIG. 1 shows a schematic perspective view of the base station antenna **100** according to the first embodiment of the present disclosure, and also shows the internal structure of the base station antenna **100**; FIG. 2 shows a schematic diagram of the base station antenna **100** without a radome **10**; FIG. 3 shows a partial exemplary diagram of the base station antenna **100'** in FIG. 1 and also shows the mounting scheme of radome supporting members **20** in the base station antenna **100**.

(24) As shown in FIG. 1, the base station antenna **100** may comprise a radio **30** (the so-called Radio module), a feed board **40**, a radiating element array **50** mounted on the feed board **40**, and the radome **10** arranged in front of the radiating element array **50**. The radiating element array **50** may comprise a plurality of radiating elements, and each radiating element extends forward from the feed board **40**. The radio **30** may feed radio frequency signals to the feed board **40** through a radio frequency connector or other conductors, and in turn feed the signals to the radiating element array **50** mounted on the feed board **40**.

(25) In order to maintain the base station antenna **100** at a lower weight and/or lower cost, in the embodiment shown in the diagram, no special reflector is disposed between the radio **30** and the feed board **40**, but the radio **30** functions as a reflector. Therefore, the radio **30** may have a metal front wall **32** that faces the rear surface of the feed board **40** and metal side walls **34** that extend forward from the metal front wall **32**. A housing recess **36** is defined by the metal front wall **32** and side walls **34** of the radio **30**. The feed board **40** and the radiating element array **50** mounted thereon may be received in the housing recess **36**. The feed board **40** may be installed on the metal front wall **32** by means of screw thread fasteners **38**, such as screws. Thus, the metal front wall **32** may be configured as the “reflector” for the radiating element array **50**, so as to reflect electromagnetic waves from the radiating element array **50** forward.

(26) In some embodiments, the metal side walls **34** of the radio **30** may extend forward from the metal front wall **32** to the front of the feed board **40**. The side circumference **12** of the radome **10** may, for example, be fixed on the extending side walls **34** of the radio **30** by means of screw thread fasteners **38**. These extending metal side walls **34** may also have functions similar to metal fences that are traditionally disposed on the reflector. In some embodiments, the size parameters of the metal side walls **34** may be adjusted so as to adjust the radiation pattern, for example, the front-to-back ratio from the radiating element array **50**.

(27) Such base station antennas **100** with the radio **30** integrated in the antenna may be the aforementioned active antenna device, which may comprise one or a plurality of radiating element arrays that operate under fifth generation (5G or later) cellular network standards. The reflector that

is generally disposed is replaced by the integrated radio **30**, for example, the metal front wall **32** thereof, thereby effectively reducing the weight and/or cost of the base station antenna **100**.

(28) In order to mount radome supporting members **20** reliably and efficiently for such reflector-free base station antennas **100**, the radome supporting members **20** may be directly fixed on the radio **30** and extend forward from the radio **30** to the radome **10** for supporting the radome **10**.

(29) As shown in FIG. 2, a plurality of rows of radome supporting members **20** arranged in a distributed manner may be mounted in the base station antenna **100**, so as to support the radome **10** fully and reliably. The corresponding radome supporting members **20**, for example, may be mounted between the radiating element array **50** to prevent the corresponding radiating elements from being damaged by the toppling of the radome **10**.

(30) Referring to FIGS. 1 and 3, the radome supporting members **20** may extend forward from the metal front wall **32** of the radio **30** to the radome **10** until they are adjacent to or abutting the radome **10**. In some embodiments, in order to mount radome supporting members **20** reliably and efficiently, each radome supporting member **20** may be integrated on the metal front wall **32** of the radio **30**—for example, through a manufacturing process such as die casting, thereby circumventing the complex mounting process of the radome supporting members **20** and saving costs.

(31) A series of through holes **42** for the radome supporting members **20** may be provided on the feed board **40**. Each radome supporting member **20** may extend forward from the front wall **32** of the radio **30**, pass through the corresponding through hole **42** and then further extend forward to the rear surface of the radome **10**. In some embodiments, the through holes **42** on the feed board **40** may be formed to conform to the radome supporting members **20** so as to effectively prevent the shaking of the radome supporting members **20** and further improve the mounting reliability of the radome supporting members **20**.

(32) In the current embodiment, the radome supporting members **20** may be constructed as support posts. It should be understood that the radome supporting members **20** may be designed in diverse forms, for example, in a cylindrical shape, a prismatic shape, a frustum shape etc., and no further restriction is made here.

(33) FIG. 4 shows a modified design scheme of the radome supporting member **20**. As shown in FIG. 4, the radome supporting member **20** may have a columnar extending section **21** extending forward from the radio **30** and a supporting end **22** for directly supporting the radome **10**.

Compared with the extending section **21**, the supporting end **22** may have an enlarged cross-section. In some embodiments, the supporting end **22** of the radome supporting member **20** may further extend forward with a gradually increasing cross-section from the extending section **21** to the supporting radome **10**. Thus, the supporting strength and reliability may be increased by increasing the contact area of the radome supporting member **20**.

(34) FIGS. 5 to 7 describe in detail a base station antenna **100'** according to a second embodiment of the present disclosure. FIG. 5 shows a schematic perspective view of the base station antenna **100'** according to the second embodiment of the present disclosure, and also shows the internal structure of the base station antenna **100'**; FIG. 6 shows a schematic perspective view of a radome **10'** of the base station antenna **100'**; FIG. 7 shows a partial exemplary diagram of the base station antenna **100'**, and shows the mounting scheme of radome supporting members **20'** in the base station antenna **100'**. It should be understood that the discussion above of the base station antenna **100** according to the first embodiment of the present disclosure may be equally applicable to the base station antenna **100** of the second embodiment of the present disclosure as long as they do not contradict each other, and it need not be described again here.

(35) The base station antenna **100'** according to the second embodiment of the present disclosure differs from the base station antenna **100** in the first embodiment in that, instead of fixing or integrating the radome supporting members **20** on the radio **30**, the radome supporting members **20'** are fixed or integrated on the radome **10'** and extend rearward from the radome **10'** to the

supporting parts of the radome supporting members 20'.

(36) As shown in FIG. 5, the radome supporting members 20' may be integrated on the rear surface of the radome 10'—for example, through a manufacturing process such as injection molding, thereby circumventing the complex mounting process of the radome supporting members 20' and saving costs. Therefore, the radome supporting members 20' may be constructed as plastic members. As shown in FIG. 5, a plurality of rows of radome supporting members 20' arranged in a distributed manner are molded on the rear surface of the radome 10' and each radome supporting member 20' may extend rearward from the radome 10' to the supporting parts for the radome supporting members 20'. It should be understood that the radome supporting members 20' may also be fixed on the rear surface of the radome 10' through feasible mounting methods, for example, screw thread connection, plugging or bonding.

(37) In some embodiments, the base station antenna 100' according to the second embodiment of the present disclosure may be a traditional passive base station antenna, for example, a 4G antenna. The radio 30' is not integrated in the passive base station antenna, but a special reflector is mounted on the passive base station antenna.

(38) In the embodiment shown in the diagram, the base station antenna 100' according to the second embodiment of the present disclosure may be realized as the aforementioned active antenna device, with the radio 30' being integrated therein. A specially disposed reflector is substituted by the integrated radio 30', for example, the metal front wall 32' thereof, thereby effectively maintaining the base station antenna 100' at a lower weight and/or lower cost.

(39) The supporting parts for the radome supporting members 20' according to the present disclosure may be disposed on any suitable position inside the base station antenna 100'. In some embodiments, the supporting parts for the radome supporting members 20' may be disposed on the feed board 40'. In some embodiments, the supporting parts for the radome supporting members 20' may be disposed on the radio 30'. In some embodiments, for example, in the case of a traditional base station antenna 100' with a reflector, the supporting parts for the radome supporting members 20' may be disposed on the reflector.

(40) In the embodiment shown in the diagram, the support part for the radome supporting members 20' may be disposed on the feed board 40'. That is to say, the radome supporting members 20' may extend rearward from the radome 10' to the supporting parts supported on the feed board 40'. As shown in FIG. 7, the supporting parts may be disposed on the screw thread fasteners 38', for example, the connecting end 39' of the screws, of the feed board 40'. The radome supporting members 20' may extend rearward from the radome 10' to the connecting end 39' abutting the screw thread fasteners 38' such that the radome supporting members 20' are stably supported on the feed board 40'. In this embodiment, the radome supporting members 20' do not need to pass through the feed board 40' again and thus may be realized with a shortened size, thereby saving materials and reducing the weight thereof. In some embodiments, the screw thread fastener 38' may be aligned with [another] screw thread fastener 38' in the forward direction, thereby saving space occupied in the base station antenna 100'.

(41) FIG. 8 shows a modified design of the radome supporting member 20'. As shown in FIG. 8, the rear end of each radome supporting member 20' may have a corresponding groove 21' for accommodating the connecting end 39' of the screw thread fastener 38', such that the rear end of the radome supporting member 20' may be spliced to the connecting end 39' of the screw thread fastener 38' in the form of shape fitting. Thus, the mounting reliability of the radome supporting members 20' may be further improved by effectively preventing the shaking of the radome supporting members 20'.

(42) Additionally, or alternately, the supporting parts for the radome supporting members 20' may be disposed on the radio 30', for example, on the metal front wall 32' thereof. To this end, a series of through holes (not shown) for the radome supporting members 20' may be provided on the feed board 40'. Each radome supporting member 20' may extend rearward from the radome 10', pass

through the corresponding through hole and then further extend rearward to the radio 30', for example, the metal front wall 32' thereof.

(43) In some embodiments, a groove (not shown) for accommodating the rear end of the radome supporting member 20' may be provided on the metal front wall 32' of the radio 30', thereby effectively prevent the shaking of the radome supporting member 20' and further improve the mounting reliability of the radome supporting member 20'.

(44) Although exemplary embodiments of the present disclosure have been described, those skilled in the art should understand that many variations and modifications are possible in the exemplary embodiments without materially departing from the spirit and scope of the present disclosure. Therefore, all variations and changes are included in the protection scope of the present disclosure defined by the claims. The present disclosure is defined by the attached claims, and equivalents of these claims are also included.

Claims

1. A base station antenna, wherein the base station antenna comprises: a radio; a feed board and a radiating element array mounted on the feed board, where the radiating element array comprises a plurality of radiating elements and each radiating element extends forward from the feed board; a radome arranged in front of the radiating element array; and a plurality of radome supporting members mounted on the radio and extending forward from the radio to the radome for supporting the radome.
2. The base station antenna according to claim 1, wherein the radio comprises a metal front wall and the feed board is mounted on the metal front wall.
3. The base station antenna according to claim 2, wherein the feed board is mounted on the metal front wall with screw thread fasteners.
4. The base station antenna according to claim 2, wherein the metal front wall is configured as a reflector for the radiating element array to reflect electromagnetic waves from the radiating element array.
5. The base station antenna according to claim 2, wherein the radio has side walls extending forward from the metal front wall.
6. The base station antenna according to claim 1, wherein through holes for the radome supporting members are provided on the feed board, and the radome supporting members extend forward from the radio, pass through the through holes and then further extend forward to the rear surface of the radome.
7. The base station antenna according to claim 1, wherein the radome supporting members are configured as support posts.
8. The base station antenna according to claim 1, wherein the radome supporting members have an extending section extending forward from the radio and a supporting end for directly supporting the radome, where the supporting end further extends forward with an enlarged cross-section from the extending section to the radome.
9. The base station antenna according to claim 1, wherein the base station antenna comprises a plurality of rows of radome supporting members.
10. The base station antenna according to claim 1, wherein the base station antenna is constructed as an active antenna.
11. The base station antenna according to claim 1, wherein the radome supporting members are plastic members or metal members.
12. A base station antenna, wherein the base station antenna comprises: a feed board; a radiating element array mounted on the feed board, where the radiating element array comprises a plurality of radiating elements, and each radiating element extends forward from the feed board; a radome arranged in front of the radiating element array; radome supporting members for supporting the

radome mounted on the radome and extending rearward from the radome; wherein the radome supporting members are integrated on the radome.

13. The base station antenna according to claim 12, wherein the radome supporting members are constructed as plastic members.

14. The base station antenna according to claim 12, wherein the radome supporting members are constructed as support posts.

15. The base station antenna according to claim 12, wherein the base station antenna comprises a plurality of rows of radome supporting members arranged in a distributed manner.

16. A base station antenna according to claim 12, wherein the base station is constructed as a 5G antenna.
